Dynamical Systems

Part II

Recorded from the lecturer in 2026-2027.

Dynamical systems – mathematical entities described by a set of equations for a set of variables w.r.t. a time-like variable.

State space or phase space.

∗ Continuous time ⇒ set of ODEs

Lotka–Volterra for rabbits (r) & sheep (s)

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad a, b, \ldots, f > 0.$$

$$\text{State space} \qquad [0, \infty) \times [0, \infty).$$

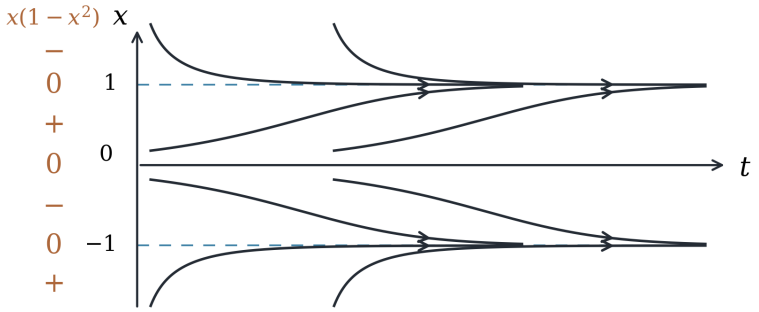
e.g. Logistic map $\qquad x_{n+1} = \mu x_n (1 - x_n), \quad 0 \le \mu \le 1$

Questions to address

- ∗ Long-term behavior of the system?
- ∗ Are there periodic solutions
- ∗ Stability of solutions, chaos
- ∗ Effect of system parameter variation

Example 1D system $\qquad \dot{x} = x(1 - x^2)$

$$\text{Solution} \qquad x(t) = \pm \left(1 + 2e^{2(t_0 - t)}\right)^{-1/2}$$



as $t \to \infty$

$x(0) > 0 \Rightarrow x(t) \to 1$

$x(0) < 0 \Rightarrow x(t) \to -1$

$x(0) = 0 \Rightarrow x(t) = 0$

Example 2D, Lotka–Volterra

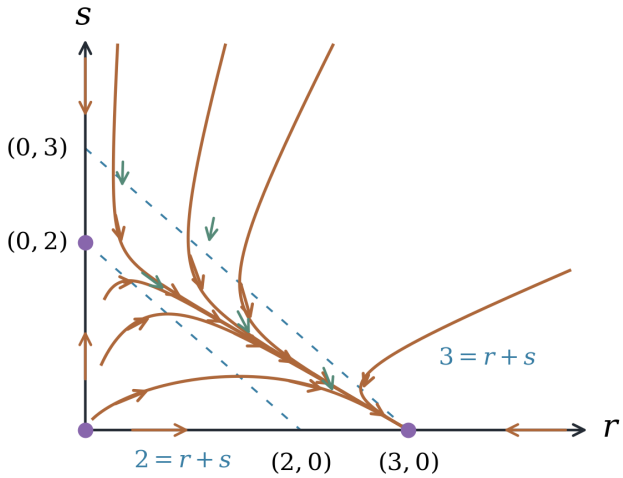
$$\dot{r} = r(3 - r - s)$$
$$\dot{s} = s(2 - r - s)$$

Can get $r(s) \Rightarrow t(s)$

∗ Find fixed points

$$\dot{r}, \dot{s} = 0 : \quad (0,0), (3,0), (0,2)$$

∗ Look at $\dot{r}, \dot{s}$ in state space



$$\text{As } t \to \infty \qquad (r,s) \to \begin{cases} (3,0), & r > 0, \\ (0,2), & r = 0,\ s > 0, \\ (0,0), & r = s = 0. \end{cases}$$

Example predator-prey

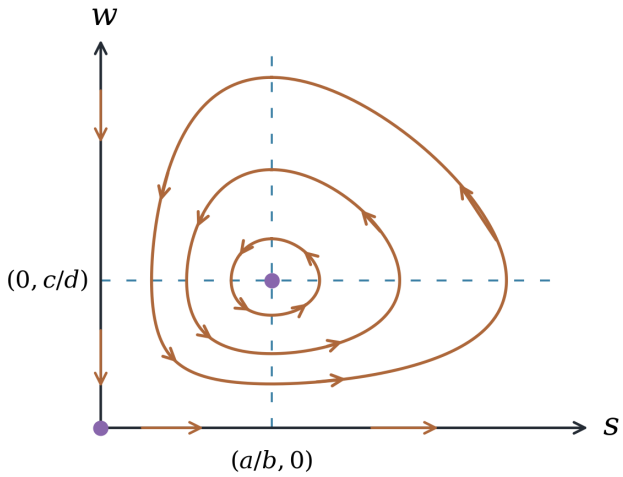
sheep (s) & wolves (w)

$$\dot{w} = w(-a + bs)$$
$$\dot{s} = s(c - dw)$$

$$a, \ldots, d > 0$$

Fixed points

$$(0,0), \quad (a/b, c/d)$$



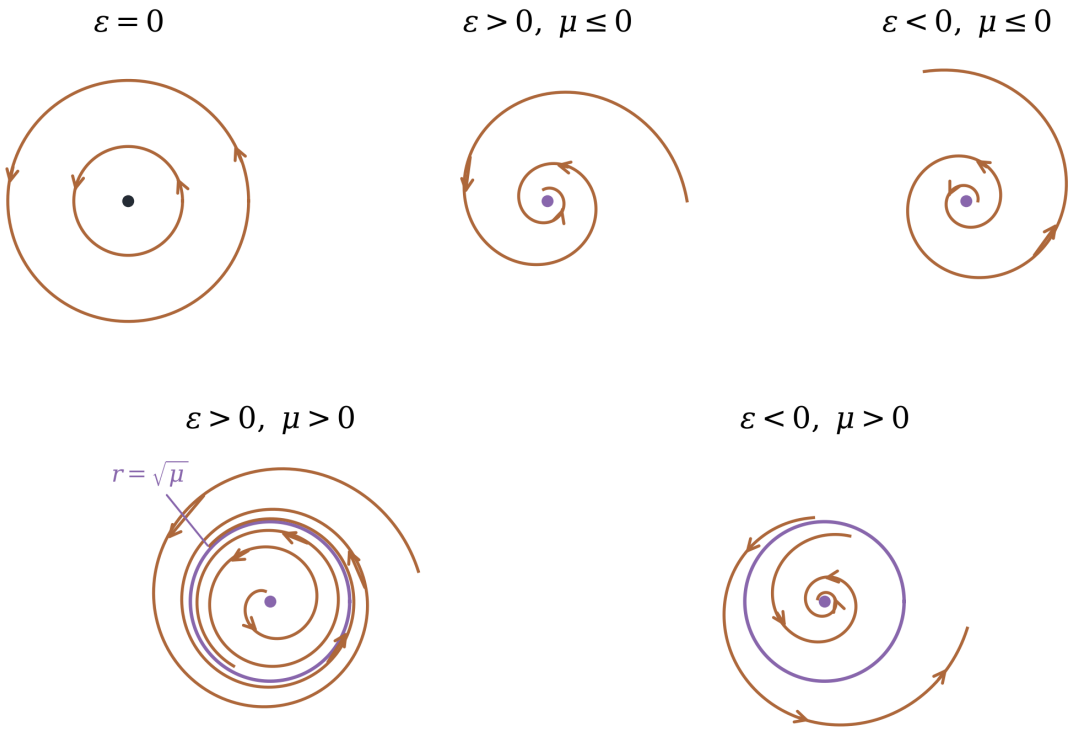
$$(s,w) \to \begin{cases} \text{periodic around } (a/b, c/d), & s > 0,\ w > 0, \\ (0,0), & s = 0,\ w \geq 0, \\ (\infty, 0), & s > 0,\ w = 0 \end{cases} \qquad \text{as } t \to \infty$$

Example

$$\dot{x} = -y + \varepsilon x(\mu - x^2 - y^2)$$
$$\dot{y} = \quad x + \varepsilon y(\mu - x^2 - y^2)$$

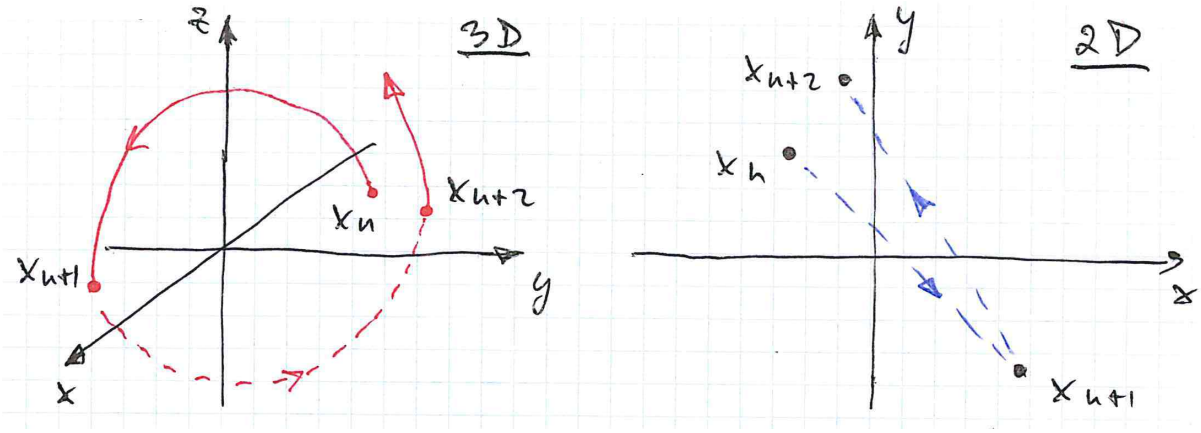
Switch to polars:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan} \frac{y}{x},$$
$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \qquad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} \quad \Rightarrow$$
$$\underline{\dot{r} = \varepsilon r(\mu - r^2)}, \qquad \underline{\dot{\theta} = 1}$$



End of Lecture 1

Poincare map in 2D for a 3D dynamical system:



Can also consider maps as discretized ODEs.

Plan for the course

(3)	§1 :	Basic definitions	ODEs
(3)	§2 :	Fixed points	
(2)	§3 :	Stability	
(5)	§4 :	Periodic orbits	
(5)	§5 :	Bifurcations	
(3)	§6 :	Maps	Maps
(3)	§7 :	Chaos	

§1. Definitions

(1.1) Define a State Space (Phase Space) $E \subseteq \mathbb{R}^n$. The state of the system is described by $\bar{x} \in E$. The state evolves in time according to an ODE of the form

$$\dot{\bar{x}} = \frac{d\bar{x}}{dt} = \bar{f}(\bar{x}, t), \qquad \text{where } \bar{f} : E \times \mathbb{R} \to E \text{ is a vector field.}$$

If $\dfrac{\partial \bar{f}}{\partial t} = 0$ the system is autonomous.

A non-autonomous system reduces to an autonomous by defining $\bar{y} = (\bar{x}, t) \in E \times \mathbb{R}$ so that

$$\dot{\bar{y}} = \begin{pmatrix} \bar{f}(\bar{y}) \\ 1 \end{pmatrix}.$$

Thus, can study only autonomous systems.

A nth order ODE

$$\frac{d^n x}{dt^n} = g\left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right)$$

can be reduced to a standard form with n equations by introducing

$$\bar{y} = \left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right) = (y_1, y_2, \ldots, y_n)$$

so that

$$\dot{\bar{y}} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \\ g(y_1, y_2, \ldots, y_n) \end{pmatrix}.$$

Similarly to autonomous systems, will only consider maps of the form

$$\bar{x}_{n+1} = \bar{F}(\bar{x}_n) \qquad \text{and not} \qquad \bar{F}(\bar{x}_n, n).$$

1.2 Initial value problem (IVP)

We would like to explore an initial value problem for $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$:

> Given the initial value of $\bar{x}$: $\bar{x}(t_0) = \bar{x}_0$, $\bar{x}_0 \in E$, $t \in I \subseteq \mathbb{R}$, find a differentiable function $\bar{x}(t)$ for $t \in I$ which remains in E for $t \in I$ and satisfies the initial condition and the ODE.

For an autonomous system can define solution in terms of a flow ϕ_t.

Definition (Flow) $\phi_t(\bar{x})$ such that $\phi_t(\bar{x}_0)$ is the solution of $\dot{\bar{x}} = \bar{f}(\bar{x})$ starting at $\bar{x}_0$ when $t = 0$ is called the flow through $\bar{x}_0$ at $t = 0$.

Thus,

$$\phi_0(\bar{x}_0) = \bar{x}_0, \qquad \phi_s\big(\phi_t(\bar{x}_0)\big) = \phi_{s+t}(\bar{x}_0).$$

Does this solution exist? Is it unique?

Theorem (Cauchy–Peano)

If $\bar{f}(\bar{x}, t)$ is continuous and $|\bar{f}| < M$ in the domain

$$D : \big\{|t - t_0| < \alpha,\ |\bar{x} - \bar{x}_0| < \beta\big\}$$

then the IVP has a solution for $|t - t_0| < \min(\alpha, \beta/M)$.

But uniqueness requires stronger conditions on f.

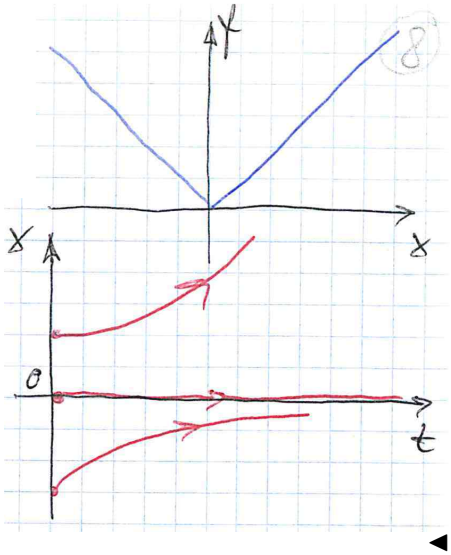
Example Unique solution

$$\dot{x} = |x|, \qquad x(t_0) = x_0$$

Solution

$$x(t) = \begin{cases} x_0 e^{t_0 - t}, & x_0 < 0, \\ 0, & x_0 = 0, \\ x_0 e^{t - t_0}, & x_0 > 0. \end{cases}$$

Note that f is continuous but not differentiable.

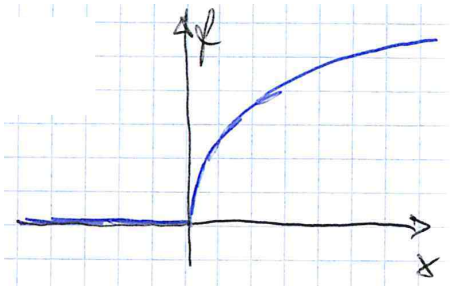


◀

Example Non-unique solution

$$\dot{x} = \begin{cases} \sqrt{x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad x(0) = 0$$

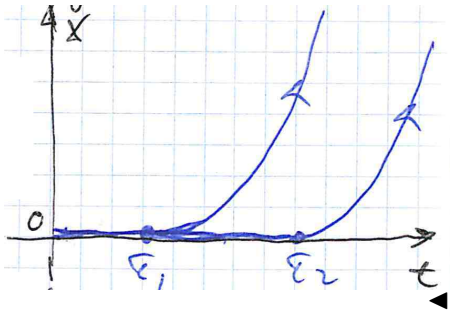
Can check that



$$x(t) = \begin{cases} 0, & t < \tau, \\ \frac{1}{4}(t - \tau)^2, & t \geq \tau \end{cases} \qquad \text{for any } \tau \geq 0$$

satisfies the equation and initial condition. But τ can be anything so a family of solutions!

Here again f is continuous and not differentiable.



◀

Why is the difference? Because $\dfrac{d}{dx}(\sqrt{x})$ is not bounded at 0!

To guarantee uniqueness, need f to be not just continuous, but also Lipschitz.

Definition Lipschitz property

A function $\bar{f}$ defined on a subset of $\mathbb{R}^n$ satisfies Lipschitz condition at a point $\bar{x}_0$ with Lipschitz constant L if $\exists\,(L, a)$ such that $\forall\,\bar{x}, \bar{y}$: $|\bar{x} - \bar{x}_0| < a$, $|\bar{y} - \bar{y}_0| < a$ we have

$$|\bar{f}(\bar{x}) - \bar{f}(\bar{y})| \leq L|\bar{x} - \bar{y}|.$$

Note that Differentiable $\rightarrow$ Lipschitz $\rightarrow$ Continuous

Theorem Uniqueness/Picard–Lindelöf/Cauchy–Lipschitz

Consider an IVP $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$ with $\bar{x} = \bar{x}_0$ at $t = t_0$. If $\bar{f}$ satisfies a Lipschitz condition at $\bar{x}_0$ then the solution $\phi_{t-t_0}(\bar{x}_0)$ exists and is unique and continuous in a neighborhood of $(\bar{x}_0, t_0)$.

However, in general uniqueness & continuity do not necessarily mean existence of solution for all time!

Example Finite time blowup

$$\dot{x} = x^2, \qquad x(0) = 1, \qquad \text{solution is } x(t) = \frac{1}{1-t},$$

blows up to ∞ at $t \rightarrow 1$.

In what follows, unless stated otherwise, we will consider only differentiable functions f.